

Dated: \_\_\_\_\_

## Normalization

### 1NF:

FD1 (Customers): customer-ID  $\rightarrow$  f-name, l-name, email, password, phone-no, created-at, house-no, street, postal-code, city, country

FD2 (Customer candidate key): email  $\rightarrow$  cust-ID, f-name, l-name, password, ph-no, created-at

FD3 (Authors): author-ID  $\rightarrow$  f-name, l-name

FD4 (Publishers): publisher-ID  $\rightarrow$  publisher-name

FD5 (Genre): genre-ID  $\rightarrow$  genre-name

FD6 (Books): book-ID  $\rightarrow$  book-name, price, rating, stock, cover-type, book-type, author-ID, publisher-ID, genre-ID

FD7 (Orders): order-ID  $\rightarrow$  customer-ID, creation-time, status, total-price

FD8 (order-items): order-ID, book-ID  $\rightarrow$  quantity, unit-price

$\rightarrow$  Since there is no multi-valued attribute in the tables, the database is already in 1NF

### 2NF:

$\rightarrow$  Only one table in the database has a composite key, the order-items table (order-ID, book-ID) and since the non-key attributes (Quantity & unit-price) is dependent on both order-ID & book-ID there are no partial dependencies and the database is in 2NF.

### 3NF:

$\rightarrow$  In FD1 customer-ID  $\rightarrow$  postal-code AND postal-code  $\rightarrow$  city, country which is a transitive dependency. To change that I would decide to create an addresses table and a countries table but since that was not in the scope of my project

128 MB

Dated: \_\_\_\_\_

being a bookstore management system & not an order-delivery management system I decided to remove postal codes from the customers table and keep only the country & city (the further details could be communicated via email or the phone number provided with the courier team).

→ hence the database is now in 3NF as well

### BCNF

→ email → customer-ID but this does not violate BCNF since email has a UNIQUE constraint and is considered a candidate key.

→ Since the rest of the tables have ~~are~~ a determinant that is a candidate / super key, the database is in BCNF